LUCIDMSK CME PROGRAM

MODULE 08 — POST-TEST

Pelvis & Hip CT: Fracture Classification Systems

Pelvis · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 Young-Burgess LC-II pelvic fracture defining feature:

A	Ipsilateral sacral compression fracture only
B	Ipsilateral sacral fracture + ipsilateral iliac crescent fracture
C	Bilateral anterior + posterior ring disruption
D	Complete SI joint disruption

2 Garden Stage III femoral neck fracture. Defining feature:

A	Incomplete; valgus impaction
B	Complete; NO displacement; trabeculae aligned
C	Complete; PARTIAL displacement; trabeculae of neck NOT aligned with acetabulum
D	Complete; full displacement; head freely rotated

3 Denis Zone II sacral fracture passes through:

A	Lateral to the sacral foramina (sacral ala)
B	The sacral foramina themselves
C	The spinal canal
D	The S1 endplate only

4 Pipkin Type II femoral head fracture:

A	Fragment inferior to fovea (non-weight-bearing)
B	Fragment SUPERIOR to fovea (weight-bearing articular surface)
C	Head fracture + femoral neck fracture
D	Head fracture + acetabular fracture

5 Pauwels Type III femoral neck fracture:

A	Horizontal (<30°); compressive; stable
B	Intermediate (30–50°); mixed
C	Vertical (>50°); SHEAR forces; UNSTABLE; highest AVN risk
D	Impacted valgus; most stable

6 Schatzker Type VI tibial plateau fracture:

A	Lateral plateau split only
B	Lateral plateau split + depression
C	Bicondylar without metaphyseal dissociation
D	Bicondylar + metaphyseal-diaphyseal DISSOCIATION

7 Judet-Letournel "T-shaped" acetabular fracture involves:

A	Anterior column only
B	Posterior column + vertical stem through obturator foramen
C	Both columns; posterior wall associated
D	Anterior column + posterior hemitransverse

8 Young-Burgess APC-III pelvic fracture. Expected injury:

A	Intact posterior SI ligaments
B	Partial SI disruption; posterior SI intact
C	Complete SI disruption; both anterior AND posterior SI ligaments torn
D	Contralateral open-book only

9 Both-column acetabular fracture pathognomonic CT finding:

A	Intact dome with anterior column fracture

B	Both column Fx; at least one segment attached to axial skeleton
C	Both column Fx; NO segment attached to axial skeleton; "spur sign"
D	Posterior wall + transverse only

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10 Posterior acetabular wall fracture involving >50% of the posterior wall after hip dislocation:

A	Stable; conservative management
B	Borderline; CT stress exam
C	Posterior hip INSTABILITY — surgical fixation of posterior wall required
D	No change from smaller fragments

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	B	C	B	B	C	D	B	C	C	C

Q1 **Young-Burgess LC-II pelvic fracture defining feature:** **Answer: B**

A Ipsilateral sacral compression fracture only

✓ **Ipsilateral sacral fracture + ipsilateral iliac crescent fracture**

C Bilateral anterior + posterior ring disruption

D Complete SI joint disruption

EXPLANATION YB LC-II = ipsilateral sacral buckle fracture + ipsilateral posterior iliac crescent (iliac wing) fracture. LC-I = sacral compression alone. LC-III = LC-II + contralateral open-book.

Q2 **Garden Stage III femoral neck fracture. Defining feature:** **Answer: C**

A Incomplete; valgus impaction

B Complete; NO displacement; trabeculae aligned

✓ **Complete; PARTIAL displacement; trabeculae of neck NOT aligned with acetabulum**

D Complete; full displacement; head freely rotated

EXPLANATION Garden III = complete fracture with PARTIAL displacement — femoral head displaced but trabeculae of neck no longer align with acetabular trabeculae. Garden IV = complete displacement with head freely rotated.

Q3 **Denis Zone II sacral fracture passes through:** **Answer: B**

A Lateral to the sacral foramina (sacral ala)

✓ **The sacral foramina themselves**

C The spinal canal

D The S1 endplate only

EXPLANATION Denis Zone II = transforaminal (through the sacral foramina). Zone I = lateral to foramina (ala). Zone III = medial (spinal canal). Zone II nerve injury rate ~28%. Zone III highest neurological risk.

Q4 **Pipkin Type II femoral head fracture:** **Answer: B**

A Fragment inferior to fovea (non-weight-bearing)

✓ **Fragment SUPERIOR to fovea (weight-bearing articular surface)**

C Head fracture + femoral neck fracture

D Head fracture + acetabular fracture

EXPLANATION Pipkin II = fragment superior to fovea capitis = weight-bearing dome involved. Type I = inferior to fovea (non-WB; better prognosis). Type III = head + femoral neck. Type IV = head + acetabular fracture.

Q5 **Pauwels Type III femoral neck fracture:** **Answer: C**

- **A** Horizontal (<30°); compressive; stable
- **B** Intermediate (30–50°); mixed
- ✓ **Vertical (>50°); SHEAR forces; UNSTABLE; highest AVN risk**
- **D** Impacted valgus; most stable

EXPLANATION Pauwels III = most vertical fracture line (>50°) — high shear stress, highest risk of non-union and AVN. Pauwels I = most horizontal (<30°), compressive forces, most stable.

Q6 **Schatzker Type VI tibial plateau fracture:** **Answer: D**

- **A** Lateral plateau split only
- **B** Lateral plateau split + depression
- **C** Bicondylar without metaphyseal dissociation
- ✓ **Bicondylar + metaphyseal-diaphyseal DISSOCIATION**

EXPLANATION Schatzker VI = bicondylar WITH metaphyseal-diaphyseal dissociation (most severe). Type V = bicondylar without dissociation. Types I–III = lateral plateau only. Type IV = medial plateau.

Q7 **Judet-Letournel "T-shaped" acetabular fracture involves:** **Answer: B**

- **A** Anterior column only
- ✓ **Posterior column + vertical stem through obturator foramen**
- **C** Both columns; posterior wall associated
- **D** Anterior column + posterior hemitransverse

EXPLANATION T-shaped = transverse fracture + vertical stem through the obturator foramen/ischiopubic region. Both-column fracture = no segment attached to axial skeleton (floating acetabulum — "spur sign").

Q8 **Young-Burgess APC-III pelvic fracture. Expected injury:** **Answer: C**

- **A** Intact posterior SI ligaments
- **B** Partial SI disruption; posterior SI intact
- ✓ **Complete SI disruption; both anterior AND posterior SI ligaments torn**
- **D** Contralateral open-book only

EXPLANATION APC-III = complete SI joint disruption — both anterior AND posterior SI ligaments torn = maximum pelvic instability. Associated with major internal iliac vessel hemorrhage.

Q9 **Both-column acetabular fracture pathognomonic CT finding:** **Answer: C**

A Intact dome with anterior column fracture

B Both column Fx; at least one segment attached to axial skeleton

✓ **Both column Fx; NO segment attached to axial skeleton; "spur sign"**

D Posterior wall + transverse only

EXPLANATION Both-column fracture = ALL articular segments detached from the axial skeleton. The pathognomonic "spur sign" = isolated fragment of iliac wing posterior to the fracture. Most common complex acetabular pattern.

Q10 **Posterior acetabular wall fracture involving >50% of the posterior wall after hip dislocation:** **Answer: C**

A Stable; conservative management

B Borderline; CT stress exam

✓ **Posterior hip INSTABILITY — surgical fixation of posterior wall required**

D No change from smaller fragments

EXPLANATION >50% posterior wall fragment = posterior hip instability after reduction. Fragments >25–40% may require fixation depending on CT stress exam. Large posterior wall fractures require ORIF to restore stability.